

Chem 0345/0330 – Organic Lab

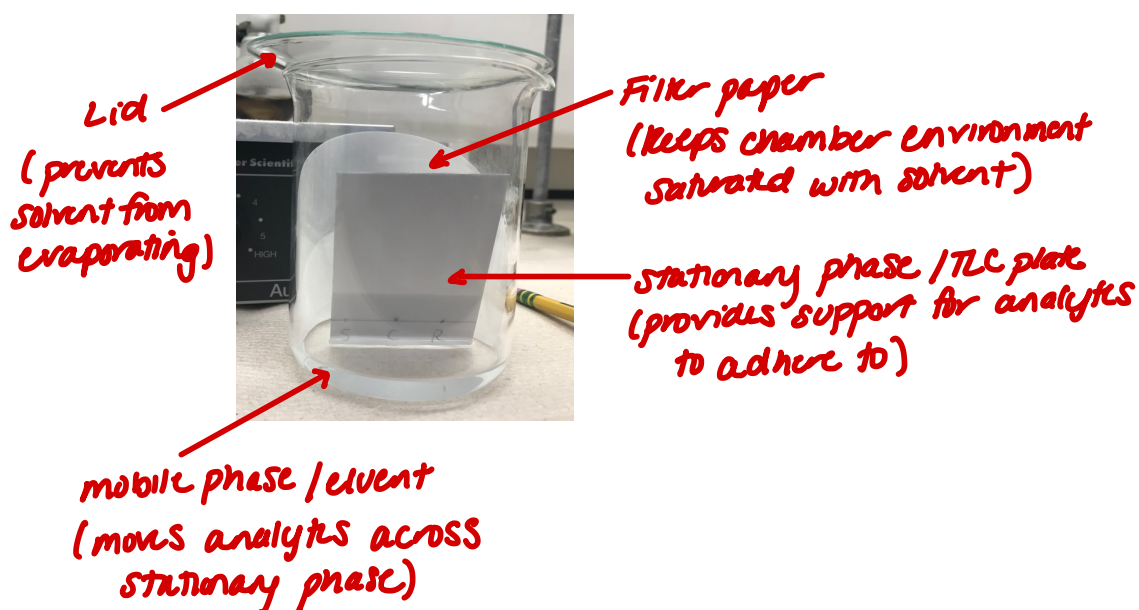
Activity 4: TLC

Extra Practice Problems

1. Classify each material below as a stationary phase or mobile phase. You may not have explicitly discussed each of the materials below; use your knowledge of how stationary and mobile phases are used in chromatography to answer this question.

Material	Stationary or Mobile Phase
Silica	<i>stationary (solid)</i>
Hexane	<i>mobile (liquid)</i>
Ethyl acetate	<i>mobile (liquid)</i>
C18-modified silica	<i>stationary (solid)</i>
Dichloromethane	<i>mobile (liquid)</i>
Alumina (Al_2O_3)	<i>stationary (solid)</i>
Methanol	<i>mobile (liquid)</i>
Diethyl ether	<i>mobile (liquid)</i>
Glass	<i>stationary (solid)</i>

2. Label the TLC setup below with each of the following components and describe each one's purpose:

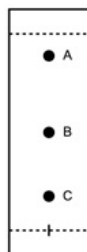


- A. Lid
- B. Filter paper
- C. Stationary phase
- D. Mobile phase

3. Briefly describe the process for running a TLC.

- 1) Draw baseline on TLC plate w/ pencil
- 2) Load TLC plate with substance using capillary tube
- 3) Put TLC plate in chamber
- 4) Monitor TLC plate
- 5) Remove plate from chamber and mark solvent line

4. What is the R_f for each spot below? What can you say about the relative polarity and retention time of each spot?



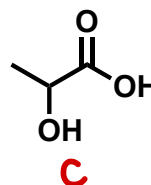
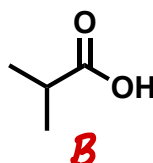
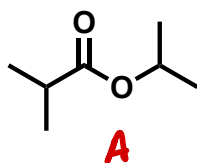
$$R_f = \frac{\text{distance spot travelled}}{\text{distance solvent travelled}}$$

$$\text{Spot C}^* = \frac{0.5 \text{ cm}}{2.6 \text{ cm}} = \boxed{0.19}$$

- A. Spot A = 0.89
 B. Spot B = 0.5
 C. Spot C = 0.19

* distances based on measurement on iPad; your measurements may be different, but ratio (i.e. R_f) is the same

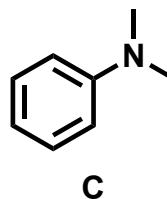
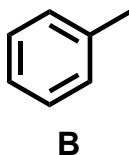
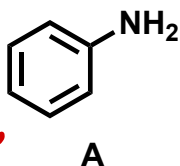
5. Assign the likely identity of each spot in Question 4 to the molecules below:



(least polar, highest R_f) (most polar due to 2x -OH, smallest R_f)

6. Which of the following statements is true about the three molecules below?

A: most polar = smallest R_f , highest retention time, largest K_p



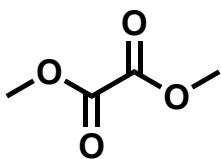
B: least polar = largest R_f , smallest retention time, smallest K_p

- ☒ A. Molecule A will have a smaller retention time than Molecule B
☒ B. Molecule A will have a larger R_f than Molecule B
☒ C. Molecule B will have a smaller retention time than Molecule C
☒ D. Molecule B will have a smaller R_f than Molecule C
☐ E. Molecule A will have a smaller K_p than Molecule B

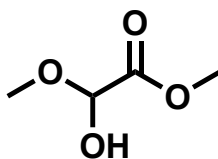
edits in purple make statement true

7. Which of the following statements is true about the three molecules below?

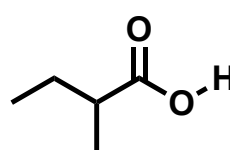
A:
least polar =
largest R_f , smallest
retention time,
smallest K_p



A



B

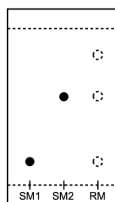


C

B:
most polar
= smallest
 R_f , highest
retention time,
largest K_p

- ☒ A. Molecule A will have a smaller retention time than Molecule B
- ☒ B. Molecule A will have a larger R_f than Molecule B
- ☒ C. Molecule B will have a smaller retention time than Molecule C
- ☒ D. Molecule A will have a smaller R_f than Molecule C
- ☒ E. Molecule A will have a smaller K_p than Molecule B

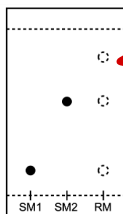
8. A reaction using starting material 1 (SM1) and starting material (SM2). After 1 hour, a TLC of the starting materials and the reaction mixture (RM) was run. Which of the following statements is true? You can assume that equal amounts of starting materials were used.



← left over SM; all should be consumed when rxn complete since there is a 1:1 ratio of starting material

- ☒ A. The reaction is complete because all starting materials have been consumed.
- ☒ B. The reaction is not complete and should be run for additional time.
- ☒ C. The reaction is complete even though all starting materials have not been consumed.
- ☒ D. The reaction is not complete because the starting materials are impure.

9. A reaction using starting material 1 (SM1) and starting material (SM2). After 1 hour, a TLC of the starting materials and the reaction mixture (RM) was run. Which of the following statements is true? You can assume that equal amounts of starting materials were used.



← product = does not have same R_f as SM1 + SM2
Has highest R_f = least polar molecule

- ☒ A. The product is more polar than both SM1 and SM2.
- ☒ B. The product is more polar than SM1 but not SM2.
- ☒ C. The product is less polar than both SM1 and SM2.
- ☒ D. The product is less polar than SM1 but not SM2.